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Non-automatable cognitive skills in higher education in the age of generative AI

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Introduction: The emergence of Generative Artificial Intelligence (GenAI) has led to an urgent reconsideration of which cognitive and metacognitive capabilities must remain central to higher education curricula. However, there is limited synthesis of the skills that cannot be meaningfully reduced to, or outsourced through, automated text generation. The current review addresses a gap in bringing together conceptual, pedagogical, and assessment-focused evidence on the human cognitive work that should be protected, developed, and evidenced in GenAI-enabled learning. This is essential for curriculum and assessment redesign in contemporary higher education contexts.

Methods: The current study presents a narrative review of a total of 24 studies examining cognitive and metacognitive skills that resist automation in higher education contexts. Searches were conducted in Scopus, ERIC, Semantic Scholar, and Google Scholar for literature published between January 2020 and December 2025.

Results: Across the studies, higher-order critical thinking emerged as the most frequently cited non-automatable skill, alongside authorial decision-making, contextual disciplinary judgment, ethical and epistemic agency, and metacognitive skills such as self-regulation and reflective thinking. Skills resisting automation were characterised by accountability, contextual reasoning, and second-order self-regulation. The literature recommended learning outcomes that emphasise higher-order thinking, AI literacy and transversal competencies, including academic integrity, collaboration, and leadership. Evidence indicated that active, collaborative and metacognitively oriented pedagogies, and scaffolded human-AI co-creation, support these skills more effectively than unguided GenAI use.

Conclusion: GenAI's impact on non-automatable skills is contingent on instructional design. Structured, reflective integration can support critical thinking and self-regulation, whereas uncritical or fully outsourced use may suppress independent analysis. Higher education should therefore redesign curricula and assessment to make accountable human thinking visible.

KEYWORDS
cognition, curriculum design, generative artificial intelligence, higher education, metacognition, non-automatable skills

Introduction

Higher education has long treated written work as evidence of cognitive engagement. An essay, a lab report, or a reflective journal was assumed to carry the marks of the thinking that produced it, because writing requires the student to actively recall, structure, and articulate what they understand. Generative AI (GenAI) has unsettled

this assumption. A well-structured and well-articulated text can now be produced in seconds without the student necessarily understanding the topic, which means that assessing only the final output increasingly amounts to assessing the tool rather than the learner (Essien, 2024; Nguyen Thanh et al., 2023).

Educators are now experimenting with ways to make student thinking visible again. One approach is to allow the use of ChatGPT in assignments and examinations, but to require students to submit the prompts they used along with the final output. This is informative because constructing a good prompt itself demands some understanding of the topic, clarity in framing the query, and the ability to interpret the system-generated response (Lee and Palmer, 2025; Raitskaya and Tikhonova, 2025). Over repeated interactions with GenAI, a thoughtful student refines prompts, asks follow-up questions, and adjusts the direction of inquiry (Buckland, 2025). This iterative process is itself a form of metacognition: an awareness of one's own understanding and its limits, and gives educators a window into not only the final answer but the reasoning behind it (Richmond and Nicholls, 2024).

This raises the importance of what we describe here as non-automatable cognitive skills in higher education. In this review, the term is not used narrowly to mean skills that AI cannot perform at all. It refers to forms of thinking that cannot be meaningfully reduced to automated text generation, or outsourced to GenAI, without losing the educational value inherent in them (Sætra, 2025). These are skills that involve human judgment in contexts where meaning, purpose, standards, and consequences matter (Raitskaya and Tikhonova, 2025).

Although existing studies have examined GenAI in relation to assessment, academic writing, critical thinking, and AI literacy, there is limited synthesis of the cognitive and metacognitive skills that should remain with the learner, the reasons why these skills resist automation, and their implications for curriculum design in higher education. This gap is important because higher education must continue to establish what a student can think, judge, and do, rather than only what a student can submit.

The present narrative review therefore brings together the available literature on non-automatable cognitive skills in higher education. Its contribution is to identify the cognitive capabilities described as non-automatable or minimally automatable, explain the characteristics that make them resistant to automation, and outline their implications for curriculum, pedagogy, and assessment redesign in the age of GenAI.

Methods

Research design

The present study adopted a narrative review. This design was selected because the topic of non-automatable cognitive skills in the age of generative artificial intelligence (GenAI) is emerging, conceptually broad, and represented by empirical studies, review papers, and conceptual discussions. A narrative review therefore enabled the study to interpret and synthesise the different perspectives on cognitive and metacognitive skills, rather than conduct an exhaustive systematic mapping of the literature.

The review focused on identifying the cognitive capabilities that remain non-automatable or minimally automatable in higher education, examining the characteristics that make these skills resistant to automation, and considering their implications for curriculum, pedagogy, and assessment redesign.

The review was guided by the following research questions:

1. Which cognitive capabilities are specifically identified in the literature as non-automatable or minimally automatable in higher education contexts?
2. What characteristics make certain cognitive and metacognitive skills resistant to automation?
3. What do these findings imply for redesigning educational practices in the age of generative AI?

Materials for the study

The materials for the study consisted of electronic databases and search engines, a set of search terms, eligibility criteria for study selection and the full-text articles included in the review. Combination of keyword searches were conducted in Scopus, ERIC, Semantic Scholar and Google Scholar for literature published between January 2020 to December 2025.

Search strategy

Because of the nature of the review, relevant literature was located through an iterative and purposive search process, rather than an exhaustive systematic protocol. The key terms entered into the search engines included: "generative artificial intelligence", "generative AI", "ChatGPT", "generative AI in higher education", "large language model", "automation", "automatable", "non-automatable", "minimally automatable", "higher education", "university", "college", "tertiary education", "cognition", and "metacognition".

Inclusion criteria

The titles and abstracts of the articles that were retrieved were first screened for relevance. Studies were eligible for inclusion if they

- were situated in a higher education context,
- explicitly examined generative AI in relation to teaching, learning, or educational practice,
- addressed cognitive or metacognitive skills, with particular attention to capacities described as non-automatable or minimally automatable.

Exclusion criteria

Peer-reviewed empirical studies, systematic, scoping, and narrative reviews, meta-analyses, and conceptual or theoretical papers were retained for further consideration. Studies focused on K-12 education, informal learning settings, or purely technical aspects of AI without educational relevance were

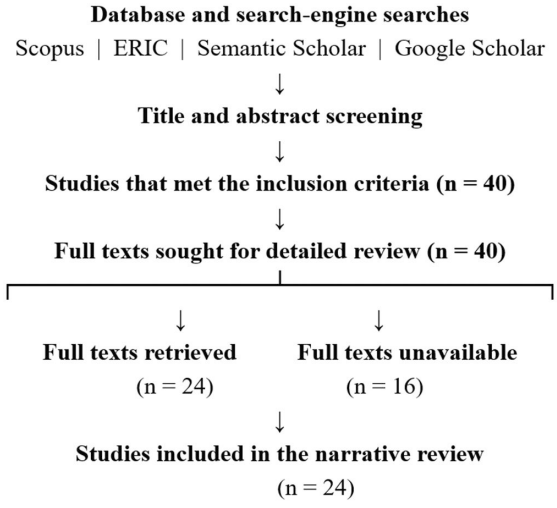


FIGURE 1
Study selection and full-text retrieval process.

excluded. Opinion pieces, editorials, and non-research commentaries lacking analytical rigor were also excluded.

Study selection

Application of these inclusion and exclusion criteria yielded a final eligible pool of forty studies, of which full texts could be retrieved for twenty-four. The remaining sixteen studies were excluded at the full-text retrieval stage because the complete versions were not available through open-access channels, or the records were limited to bibliographic metadata. Therefore, they could not be reviewed with the depth required for analytical synthesis. The present narrative review is therefore based on these twenty-four full-text papers, which were read in their entirety and analysed thematically to address the guiding research questions. The study selection and full-text retrieval process is shown in Figure 1.

Results

Characteristics of included studies

This narrative review includes 24 peer-reviewed studies published between January 2020 and December 2025, with the vast majority published in the most recent two years, reflecting the rapid scholarly response to generative AI developments in higher education. The characteristics of the included studies, including study type, educational context, and primary focus area, are summarised in Table 1.

Thirteen are secondary syntheses i.e., systematic, scoping, or narrative reviews that map the field at the level of evidence (e.g., Deroncele-Acosta et al., 2025; Komasawa and Yokohira, 2025 Lee and Palmer, 2025; Qian, 2025; Raitskaya and Tikhonova, 2025; Tillmanns et al., 2025 Weng et al., 2024; Xia et al., 2024). Seven are empirical primary studies, including quasi-experimental designs, capability evaluations, and curricular case studies (Buckland and Purcell, 2025; Essien et al., 2024; Arif and Naeem, 2025; Lindbäck et al., 2025; Michalon and Camacho-Zúñiga, 2023; Nguyen Thanh et al., 2023; Richmond and Nicholls, 2024). The remaining four are conceptual or position pieces that develop frameworks, faculty-perception analyses, or process-oriented critiques (Buele and Llerena-Aguirre, 2025; Francis et al., 2025; Gonsalves, 2024; Sætra, 2025).

Geographically, the research spans multiple continents including Asia (Malaysia, Vietnam), Europe (UK, Sweden), North America, Australia, and the Philippines, with several studies adopting international or cross-cultural perspectives. Across studies, disciplinary contexts also vary.

Narrative synthesis

This narrative review examines the literature through three guiding questions concerning the nature of non-automatable cognitive capabilities in higher education.

Conceptual grounding of Non-automatable cognitive skills

What does it actually mean to call a cognitive skill “non-automatable"? The literature uses the term in three distinguishable senses. In one usage, non-automatability is understood as a property of task complexity: GenAI competence starts degrading as we go higher up the Bloom’s taxonomy. So, evaluation and creation are non-automatable by default (Essien, 2024; Gonsalves, 2024; Nguyen Thanh et al., 2023). In a second usage, the term could be understood as a property of regulation, as in, even when AI can produce a competent output, only a learner can monitor, justify, and revise their own reasoning. The absence of this second-order activity is precisely what GenAI is unable supply (Arif and Naeem, 2025; Li, 2024; Raitskaya and Tikhonova, 2025; Tillmanns et al., 2025). In a third, the term carries a certain normative weight. It implies that certain capacities such as authorship, judgment under uncertainty, professional accountability, should not be delegated because their delegation undoes the evidentiary value it holds as a proof of the person’s competence (Buele and Llerena-Aguirre, 2025; Deroncele-Acosta et al., 2025; García-López and Trujillo-Liñán, 2025; Komasawa and Yokohira, 2025; Sætra, 2025). Taken together, we can understand that higher-order tasks tend also to be the tasks that require self-regulation, and the tasks that require self-regulation tend to be the ones that carry epistemic responsibility. The construct this review adopts therefore treats non-automatable cognition as constituted not by what AI cannot produce but by who is accountable for, regulates, and authorises the cognitive work.

Cognitive skills identified as non-automatable or minimally automatable

The skill that appears most often in the reviewed studies is higher-order critical thinking. This includes evaluation, synthesis, and informed judgment. Studies across business education (Essien, 2024), economics, marketing (Gonsalves, 2024), and language-related disciplines (Raitskaya and

TABLE 1 Summaries of reviewed studies.

Sr. no.	Author(s) (year)	Study type	Educational context	Primary focus area
1	Michalon and Camacho-Zúñiga (2023)	Empirical case study	Industrial engineering	Curriculum/transversal competencies
2	Nguyen Thanh et al. (2023)	Empirical study	Economics/business	Authentic assessment
3	Essien et al. (2024)	Empirical study	UK business schools	Critical thinking across Bloom's levels
4	Gonsalves (2024)	Conceptual/framework	Marketing education	Curriculum/Bloom's revision
5	Richmond and Nicholls (2024)	Empirical case study	Undergraduate psychology	Assessment & AI literacy
6	Weng et al. (2024)	Scoping review	Higher education (cross-disciplinary)	Assessment & learning outcomes
7	Xia et al. (2024)	Scoping review	Higher-education assessment	Assessment transformation
8	Arif and Naeem (2025)	Empirical study	EFL writing	Academic writing & language learning
9	Buckland and Purcell (2025)	Empirical case study	Organisational behaviour	Assessment redesign
10	Buele and Llerena-Aguirre (2025)	Integrative review	Higher education (cross-disciplinary)	Faculty roles & academic work
11	Chanpradit (2025)	Systematic review	Academic writing	Academic writing
12	Deep and Chen (2025)	Systematic review	Academic writing	Academic writing
13	Deroncele-Acosta et al. (2025)	Systematic review	Higher education (transversal skills)	Curriculum/competencies
14	Francis et al. (2025)	Conceptual/position paper	Biomedical sciences	Ethics & integrity framework
15	García-López and Trujillo-Liñán (2025)	Systematic review	Higher education (ethics/regulation)	Ethics & regulation
16	Gundu (2025)	Conceptual/framework	Higher-education e-assessment	Assessment design
17	Komasawa and Yokohira (2025)	Narrative review	Medical education	Professional/clinical education
18	Lee and Palmer (2025)	Systematic review	Higher education (prompt engineering)	Curriculum/AI literacy
19	Li and Balinas (2025)	Systematic review	Information-literacy education	Information literacy
20	Lindbäck et al. (2025)	Empirical case study	Physiotherapy education	Professional/clinical education
21	Qian (2025)	Systematic review	Higher education (pedagogy)	Pedagogical applications
22	Raitskaya and Tikhonova (2025)	Scoping review	Language/cross-disciplinary	AI-human interaction & critical thinking
23	Sætra (2025)	Conceptual essay	Research training	Research/scholarly process
24	Tillmanns et al. (2025)	Systematic review	Higher education (cross-disciplinary)	Curriculum & learning spaces

Studies ordered chronologically by year, then alphabetically by first author surname within each year.

Tikhonova 2025) report a similar pattern. GenAI tools work well with lower to medium levels of Bloom's Taxonomy.

A second skill is authorial decision-making in academic writing. Several authors argue that writing is not just a way of expressing ideas; it is the process through which thinking takes shape (Chanpradit, 2025; Deep and Chen, 2024; Michalon and Camacho-Zuñiga, 2023). When students use GenAI to produce a draft, the final product may look good, but the thinking process that usually accompanies writing is missed (Sætra, 2025). These include decisions such as what to keep, what to change, and on what grounds (Arif and Naeem, 2025; Richmond and Nicholls, 2024).

A third skill is contextual disciplinary judgment. This refers to the ability to apply knowledge to specific situations, such as a doctor treating a particular patient or a teacher adjusting to a particular class. Komasawa and Yokohira (2025) discuss this in the context of medical education. They argue that the main limitation of GenAI in clinical training is not recall of medical facts but clinical reasoning, which integrates incomplete information with patient context. Buele and Llerena-Aguirre (2025) and Qian (2025) extend this argument across disciplines. They point out that disciplinary knowledge has tacit aspects that the structured outputs of GenAI cannot capture.

Metacognitive skills resistant to automation

If critical thinking and authorial judgment describe what students reason about, metacognition describes how they manage that reasoning. In fact, several authors place metacognition at the centre of non-automatable cognition (Li, 2024; Raitskaya and Tikhonova, 2025; Tillmanns et al., 2025; Xia et al., 2024). A GenAI tool can produce text that looks like the learner has reflected upon the topic and then written it. But it cannot actually reflect on the student's own thinking, because that thinking belongs to the student, not to the model.

This has a practical effect in the classroom. When a student uses the AI output as is, without questioning it, it's not just the quality that gets hampered. The student misses out on the entire metacognitive activity—noticing what they do not understand, identifying gaps, deciding what to check, and updating their mental model (Arif and Naeem, 2025; Deroncele-Acosta et al., 2025). This activity does not happen automatically just because a task asks for it. Richmond and Nicholls (2024) found that even when undergraduate psychology students were specifically asked to compare AI-generated text with a published study, some of them still failed to spot the inaccuracies.

Characteristics of cognitive and metacognitive skills that make them resistant to automation

Why do these specific skills resist automation? The reviewed papers point to three reasons. The first is accountability. Clinical reasoning (Komasawa and Yokohira, 2025), professional ethics (Deroncele-Acosta et al., 2025; García-López and Trujillo-Liñán, 2025), and the voice of a scholarly author (Sætra, 2025) are repeatedly mentioned as examples. AI output may be reasonable, but AI cannot be held responsible for it in the way that a human being can.

The second reason is context-dependence. Disciplinary reasoning depends on factors such tacit knowledge, situational judgment, and lived experience. GenAI cannot capture these at the level of detail its outputs sometimes seem to suggest (Buele and Llerena-Aguirre, 2025; Michalon and Camacho-Zuñiga, 2023; Qian, 2025). Reading a situation correctly, knowing which features are important, which background knowledge applies, and where the limits of the problem lie, is a kind of work that AI cannot do for the learner.

The third reason is second-order self-regulation. Monitoring one's own thinking is, by its nature, something only the thinker can do (Arif and Naeem, 2025; Raitskaya and Tikhonova, 2025). AI can suggest changes to a draft, but only the author can judge whether those changes make the argument better, in alignment with their own goals.

Risks of over-reliance and skill atrophy

The reviewed studies highlight two main concerns. The first is cognitive offloading, where students hand over the effort of thinking to the tool. The second is automation bias, where students accept AI output without sufficient critical examination (Essien, 2024; Li, 2024). The more careful authors treat these risks as conditional, not automatic. The risk is not whether GenAI causes skill loss in general, but under which classroom and assessment conditions skill loss becomes likely (García-López and Trujillo-Liñán, 2025; Sætra, 2025).

The evidence is consistent with this conditional view. In unscaffolded classroom conditions, students show poorer source evaluation, weaker argument depth, and less independent judgment (Chanpradit, 2025; Deep and Chen, 2024). In scaffolded conditions that require students to critique, revise, and justify the AI's output, similar students can be seen retaining or even improving their higher-order reasoning (Arif and Naeem, 2025; Raitskaya and Tikhonova, 2025; Richmond and Nicholls, 2024). There are risks, however, they depend strongly on the design of the learning environment. The educational response, therefore, is not to ban AI but to design the conditions in which students continue to use these cognitive skills.

Recommendations for curricula and assessment redesign

At the curriculum level, AI literacy should be treated not as a separate module on "how to use the tool" but as an integrated competency. It should combine critical evaluation of AI output, ethical reasoning about its use, and awareness of one's own dependence on it (Francis et al., 2025; Gonsalves, 2024; Lee and Palmer, 2025; Li, 2024). Several authors recommend revising Bloom-style frameworks so that the higher levels clearly include higher-order cognitive skills that GenAI cannot perform (Gonsalves, 2024; Nguyen Thanh et al., 2023). Prompt engineering is useful here not because writing prompts is itself a transferable skill, but because writing good prompts forces students to externalise their own disciplinary reasoning.

At the assessment level, across creative coursework (Buckland and Purcell, 2025), physiotherapy training (Michalon and Camacho-Zuñiga, 2023), psychology education (Richmond and Nicholls, 2024), and AI-literacy task design (Lindbäck et al., 2025), the same design features come up: authentic tasks, open-ended problems, visibility of the process, and built-in self-assessment

Nguyen Thanh et al. (2023). provide a useful method for redesigning assessment: test GenAI directly on the existing assessment, find out where it fails, and then redesign assessment around those points where human cognition is still required. Komasawa and Yokohira (2025) apply the same idea in medical education through tools like OSCEs, portfolios, reflective writing, and case-based discussions.

The overall position in the literature is constructive and not restrictive. Instead of building barriers around AI use, the studies recommend building tasks that need the kind of cognition the degree is meant to certify (Deroncele-Acosta et al., 2025; Qian, 2025; Tillmanns et al., 2025).

Discussion

The dividing line all the reviewed papers identify between automatable and non-automatable cognition is similar in each case. This convergence suggests that the boundary is a property of how generative models behave under tasks requiring accountable judgment, rather than a feature of any one curriculum (Francis et al., 2025; Sætra, 2025).

Across the studies, non-automatable cognition is less a claim about the permanent limits of GenAI than an account of the work for which students must remain responsible. Critical thinking, metacognition, authorial judgment, and contextual reasoning matter because they allow a learner to evaluate an answer, recognize its limits, and explain why a particular decision has been made. This is exactly why a fluent or technically correct AI-generated response cannot, on its own, demonstrate learning. In academic writing, the important question is not only whether a student can produce a coherent text, but also whether the student has selected, evaluated, and revised ideas for defensible reasons. The same issue appears in professional education, where an answer must be judged in relation to a particular case rather than accepted simply because it appears plausible (Komasawa and Yokohira, 2025; Michalon and Camacho-Zúñiga, 2023; Sætra, 2025).

Reported effects of GenAI on higher-order skills are inconsistent across studies. The split, however, maps cleanly onto how GenAI is used in the classroom. Risks are reported where students use GenAI as a substitute for their reasoning

(Chanpradit, 2025; Deep and Chen, 2024; Essien, 2024). Gains are reported where students are required to critique, revise, or justify AI output (Arif and Naeem, 2025; Richmond and Nicholls, 2024; Tillmanns et al., 2025). So, the outcomes are predicted by the structure of the task, not by access to the tool. This reframes the policy debate from access to scaffolding standards.

The idea of an 'accountable learner-in-context' offers a way to use these findings rather than only describe them. Its three features: accountability, context-dependence, and second-order self-regulation explain why the cognitive boundary observed across disciplines holds. This also has implications in what classroom scaffolding around GenAI must protect (Buele and Llerena-Aguirre, 2025; Qian, 2025; Sætra, 2025).

The studies also point towards tasks that retain some record of the process involved in thinking, a shift away from product-only assessment, towards assessment formats that make the student's thinking visible. These include iterative drafts, prompt logs, viva voce, rubric-based critiques of AI output, and case-based tasks (Gundu, 2025; Weng et al., 2024; Xia et al., 2024).

Limitations of the study

The current narrative review acknowledges significant limitations. A few studies that were reviewed in the present study are preliminary or exploratory, or based on small samples, short interventions, short-term observations and self-reported data. Few studies have looked into long-term retention or transfer of skills after AI exposure. The research also tends to be concentrated in certain regions and academic fields, which limit generalizability of findings. Measurement of skills like critical thinking also varies across studies, making comparisons difficult. There are ongoing debates concerning edge cases (for example, whether generative AI can ever teach values or creativity directly), which the current synthesis has not been able to shed light onto, Thus, the synthesis presented here is provisional and should be validated by further research.

Future research directions

Future studies may include:

- Longitudinal investigations tracking skill development with and without AI integration.
- Large-scale experiments comparing traditional vs AI-augmented instruction on learning outcomes.
- Cross-disciplinary research to see how the usage of AI has an impact across different fields (e.g., humanities vs. sciences).
- Specific pedagogical interventions (for instance, explicit metacognitive training or collaborative AI tasks).

Conclusion

To the first guiding research question, the reviewed literature points to five main clusters of non-automatable cognitive capacity: higher-order critical thinking, authorial decision-making, metacognition, ethical and epistemic agency, and contextual disciplinary judgment.

To the second question, the common feature across all five is that each describes a property of the accountable learner-in-context. These skills carry consequences, depend on situated knowledge, and require second-order self-regulation. For these reasons they cannot be performed by a substitute.

To the third question, the educational redesign that follows is structural in nature. Curriculum should treat AI literacy as a higher-order integrated competency that combines critical evaluation, ethical reasoning, and metacognitive awareness. Assessment should shift clearly from product-based to process-based forms, using iterative drafts, prompt logs, rubric-anchored critique of AI output, oral defences, and authentic case-based.

Writing pedagogy should treat GenAI as a partner whose output students must evaluate, modify, and justify, so that writing remains the space in which thinking is formed. Institutions should reaffirm scholarship as a process and develop critical AI awareness in faculty. The value of higher education in this period lies in the cognitive work that cannot responsibly be delegated. The pedagogical task is to design the conditions in which this work remains visible, effortful, and accountably human.

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SJ: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Visualization, Writing – original draft, Writing – review & editing.

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Generative AI statement

The author(s) declared that generative AI was not used in the creation of this manuscript.

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